

SAI VARUN GADDE

+1 (469) 322-9457 | gaddevarun111@gmail.com | [Linkedin](#) | [Github](#)

SUMMARY

- * Data Engineer with **5+ years** designing large-scale ETL/ELT pipelines, Lakehouse architectures, and cloud-native data platforms across **Azure (Databricks)** and **AWS (Snowflake)**.
- * Strong in **Python, SQL, Spark, dbt**, and building validated, test-driven pipelines using reconciliation checks, drift detection, anomaly scoring, and profiling techniques.
- * Hands-on experience with **Databricks, Snowflake, Redshift, Synapse, ADF, Glue**, and workflow orchestrators such as **Airflow** and cloud-native schedulers.
- * Skilled in **dimensional modeling, SCD Type 2, semantic layer design, query tuning**, and warehouse performance optimisation.
- * Known for delivering data systems that are **fast, reliable, observable, and provably accurate** through strong engineering principles and measurable validation.

TECHNICAL SKILLS

- **Programming:** Python, SQL, PySpark, Java, Bash
- **Azure & Databricks:** Azure Databricks, Delta Lake, ADF, Synapse, ADLS, Azure DevOps
- **AWS & Snowflake:** AWS Glue, Lambda, Step Functions, S3, Redshift, Snowflake (Snowpipe, Streams, Tasks, RBAC)
- **Big Data & Compute:** Apache Spark, Spark SQL, Hadoop, Hive, AQE, Partition Optimization
- **Modeling & ELT:** dbt (tests, macros, models), Star/Snowflake schema, SCD Type 1/2, Semantic Layers
- **Data Quality & Validation:** Reconciliation, Drift Detection, Schema Validation, Great Expectations (basic), Anomaly Detection
- **Orchestration:** Airflow, Databricks Workflows, ADF Pipelines, Step Functions
- **CI/CD & DevOps:** Azure DevOps, GitHub Actions, Jenkins, Git, Terraform
- **Streaming:** Kafka, Kinesis, Event Hubs, CDC
- **Storage & Formats:** Parquet, ORC, Delta Lake, JSON, Avro
- **BI Tools:** Power BI, Tableau
- **Governance & Security:** RBAC, Key Vault, Secrets Manager, GDPR/HIPAA-aware designs

EXPERIENCE

PNC Bank

Jan 2024 - Present

Data Engineer

USA

- Implemented scalable ETL/ELT pipelines using Azure Databricks and ADF, ensuring reliable processing of 100TB+ financial data monthly with uninterrupted availability.
- Designed dimensional models (star, SCD2) to support fraud detection, risk scoring, and operational analytics across the enterprise.
- Built modular dbt models with 60+ automated tests (schema, relationships, uniqueness), increasing data trust by 45%.
- Implemented robust data quality frameworks including drift detection, reconciliation, profiling, and anomaly scoring, reducing data defects by 35%.
- Tuned Spark performance (partition pruning, caching, broadcast joins, file compaction) achieving 40% faster pipelines.
- Developed near-real-time ingestion pipelines reducing reporting latency from 24 hours to under 1 hour.
- Partnered with Finance, Risk, and Compliance teams to align KPI logic, business rules, and reporting models.
- Automated CI/CD through Azure DevOps, supporting version-controlled deployments for Databricks, Synapse, and dbt assets.
- Created reusable PySpark utilities for ingestion, validation, and error-handling, improving team productivity by 25%.
- Delivered high-quality feature datasets for ML teams, improving fraud prediction and forecasting model accuracy by 12–18%.
- Delivered data engineering solutions in Agile environments, aligning sprint goals with business priorities and ensuring continuous integration, testing, and incremental improvements.

Optum

Aug 2021 - Dec 2022

Data Engineer

- Built scalable ingestion pipelines using AWS Glue (Spark) processing 5M+ clinical & claims records daily, achieving 99.8% pipeline reliability.
- Designed Snowflake marts using SCD2, dimensional modeling, clustering, powering 20+ clinical operations dashboards.
- Implemented Snowpipe + Streams + Tasks for near real-time ingestion, reducing data latency from 6 hours to <10 minutes.
- Developed validation workflows (referential integrity, completeness, anomaly detection) decreasing QA escalations by 40%.
- Created curated analytic layers supporting care-path insights, patient outcomes, and utilization analysis.
- Tuned Snowflake performance, achieving 2–3× faster execution with 25% compute cost savings.
- Designed ML-ready datasets powering risk scoring and anomaly detection models used by data science teams.
- Automated CI/CD with GitHub Actions for SQL migrations and Snowflake object deployments.
- Developed KPI dashboards in Power BI providing unified visibility to 30+ clinicians and stakeholders.
- Strengthened ingestion resiliency with schema evolution logic, DLQ routing, and retry mechanisms.

Tech Mahindra

Jul 2020 - Jun 2021

Data Engineer

- Engineered ETL pipelines using AWS Glue, PySpark, Lambda, ingesting 500,000+ behavioral and marketing events per day.
- Designed Snowflake models supporting segmentation, funnel analytics, attribution, and customer behavior insights for marketing teams.
- Built Snowpipe ingestion achieving 95% reduction in latency, enabling near-real time dashboard refreshes.
- Implemented DQ workflows (reconciliation, drift detection, anomaly scoring), improving dataset accuracy by 40%.
- Tuned Snowflake queries delivering 2–3× performance gains and 30% compute cost reduction.
- Created ML-ready features used in churn prediction and recommendation systems, increasing model training efficiency by 25%.
- Automated BI/reporting pipelines reducing manual reporting effort by 50%.
- Built Power BI dashboards consumed by 30+ business and product leaders.
- Designed metadata-driven ingestion enabling zero-touch schema evolution and scalable transformations.
- Improved pipeline reliability by implementing idempotent loads, retry logic, and DLQ workflows, reducing failures by 70%.

EDUCATION

University of North Texas

Master of Science, Computer Science